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Information Retrieval (Master's Level)
5LN712

Retrieval-Augmented Generation with TinyLlama

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What is TinyLlama?

- A **Small Language Model (SLM)** developed by Nguyen et al. (2024)
- Trained on **~1 billion parameters** and **3 trillion tokens** of textual data
- Demonstrates good results on problem solving and some language comprehension tasks compared to SLMs of similar sizes (Nguyen et al., 2024)
- Accessible and easy to use, due to its compact size and streamlined setup process



What is Retrieval-Augmented Generation (RAG)?

- Aims to enhance the **reliability** of **generative language models**.
- Combines the **existing knowledge of the language model** (parameterised knowledge) with the **knowledge of an external source** (non-parameterised knowledge).
- Generative language models can create falsehoods, and **RAG ideally ‘anchors’ a generated response** within the **context of an information retrieval system**.

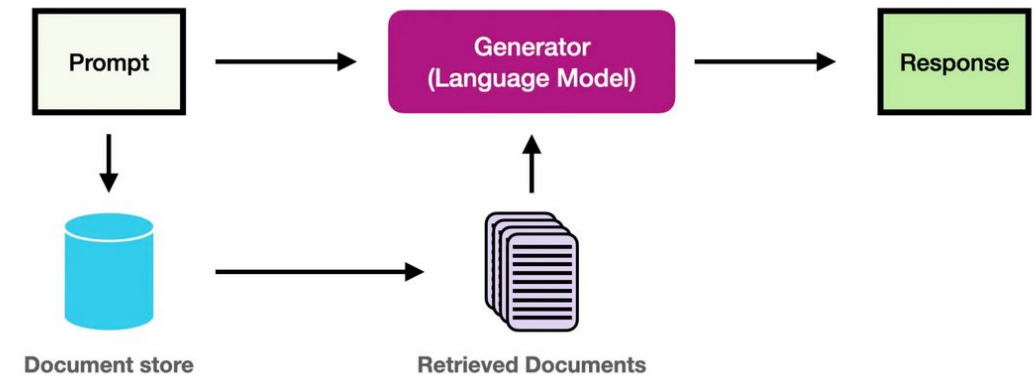


Figure 1. Example RAG Pipeline



Project Aims

- Established work on RAG and Large Language Models (LLMs), but less work on how it improves Small Language Models like TinyLlama.
- We are interested in how well TinyLlama works in an information retrieval setting

Our key research questions:

1. **Document retrieval:** How does a **RAG document retrieval pipeline with SBERT and TinyLlama** compare to a **simple baseline**?
2. **Answer generation:** How does TinyLlama's **one-shot answer generation performance** compare to that of its **zero-shot performance**?



Our Dataset

- **Competitive Undergraduate Course Question-Answer Dataset** by authors and students at Carnegie Mellon University and the University of Pittsburgh, hosted at **Kaggle**
- **150 files** with text from Wikipedia articles (n words = 690,000), and a set of **question-answer pairs**, with **at most one relevant article file** linked to each question-answer pair. Topics from the **general domain** (e.g., Presidents, Cities, Artists)
- For our experiments, we sampled a **subset of 8 topics**, one topic from each thematic group (see table 1), **172 queries** in total

General Topic	Specific Topic
Animals	Kangaroo
Countries	Liechtenstein
US Presidents	John Adams
Scientists	Blaise Pascal
Musical Instruments	Piano
Cities	London
Languages	English Language
Artists	Pablo Picasso

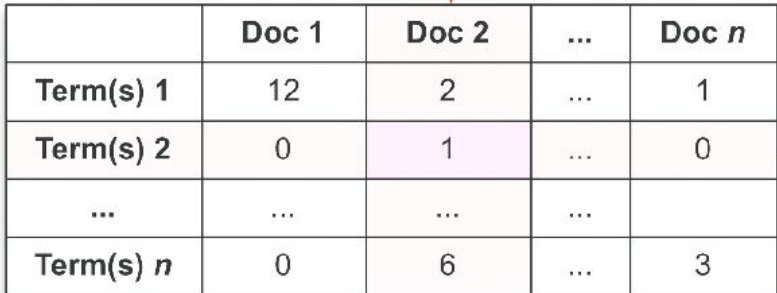
Table 1. Our selected topics for RAG experiments



Document Retrieval Baseline: TF-IDF pipeline

- Each text document had a vector of term frequency-inverse document frequency (TF-IDF) calculations
- Query converted to TF-IDF vector
- **Cosine similarity score** measured between the document and query vector
- Documents with highest cosine similarity scores between their TF-IDF vector and the query's vector considered the most relevant

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \times \text{idf}(t, D)$$



	Doc 1	Doc 2	...	Doc n
Term(s) 1	12	2	...	1
Term(s) 2	0	1	...	0
...	...	...	...	...
Term(s) n	0	6	...	3

Figure 2. Illustration of TF-IDF calculation matrix

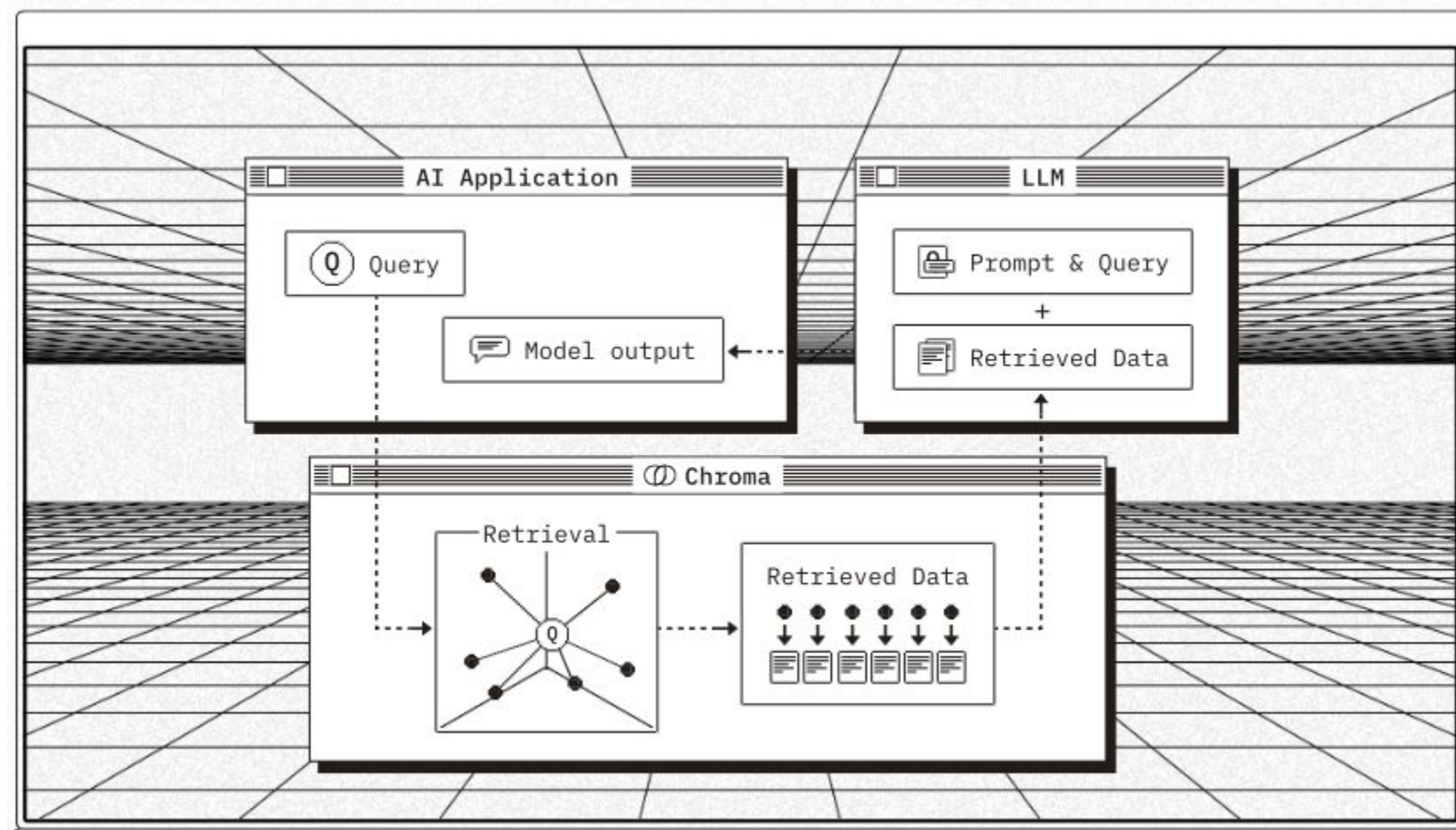
Performance:

Precision@1: 83.4%

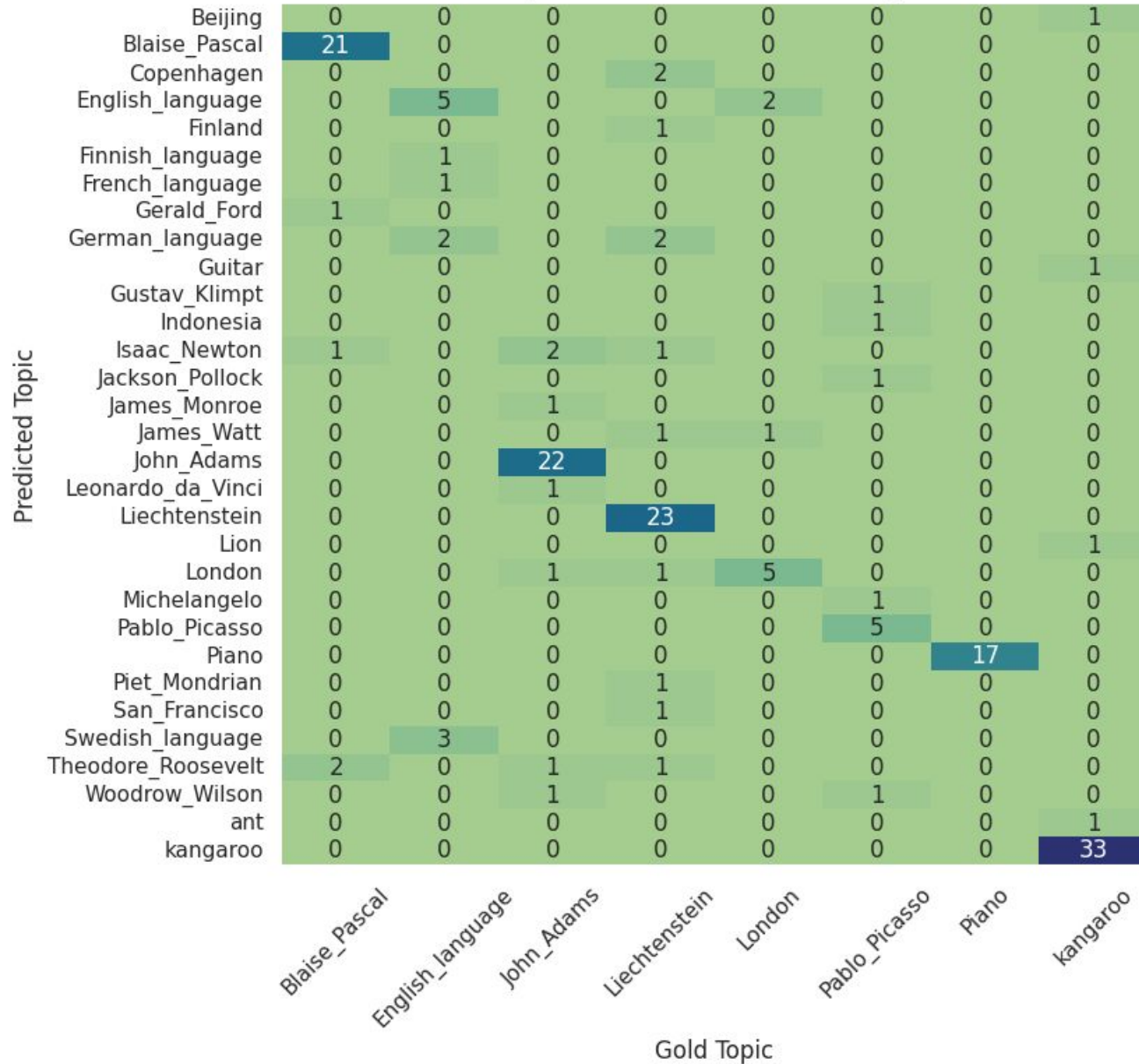
Precision@2: 87.2%



(Classic) RAG with a Database: Pipeline



Topic Confusion Matrix: Precision@1



Retrieval: Chroma

`collection.query(string)`

- SBERT embedding
- Returns top-k semantic matches

`collection.get(database_item)`

- Returns the specified database-item, ex. 'Metadata' for the path/filename

Evaluation

- 131/172 files correctly retrieved = 76%
- Manual error analysis



Answer Generation: TinyLlama

- TinyLlama loaded directly from Huggingface
- Input query and the retrieved document(s) into the prompt template
- Added memory step for previous query-document as context



Evaluation:

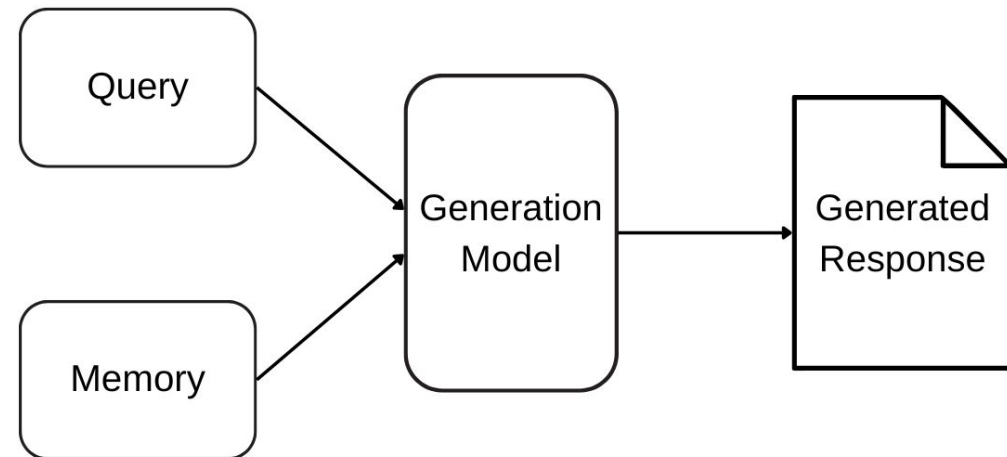
- BERTScore per Answer/Reference document pair

Topic	Precision	Recall	F1	Total questions
John_Adams	0.8160050511	0.6826085736	0.742200576	29
kangaroo	0.8279631702	0.695215019	0.7550062312	37
Liechtenstein	0.8248901595	0.6684503082	0.7370054669	34
Blaise_Pascal	0.803822937	0.6616752076	0.7248131394	25
English_language	0.8341032912	0.7043230434	0.7632175485	12
London	0.7998314723	0.6928406358	0.7419527769	8
Pablo_Picasso	0.8071025848	0.6808703721	0.7375179648	10
Piano	0.8399666408	0.7229148465	0.7763694349	17
Mean Values	0.8192106634	0.6886122508	0.7472603923	172



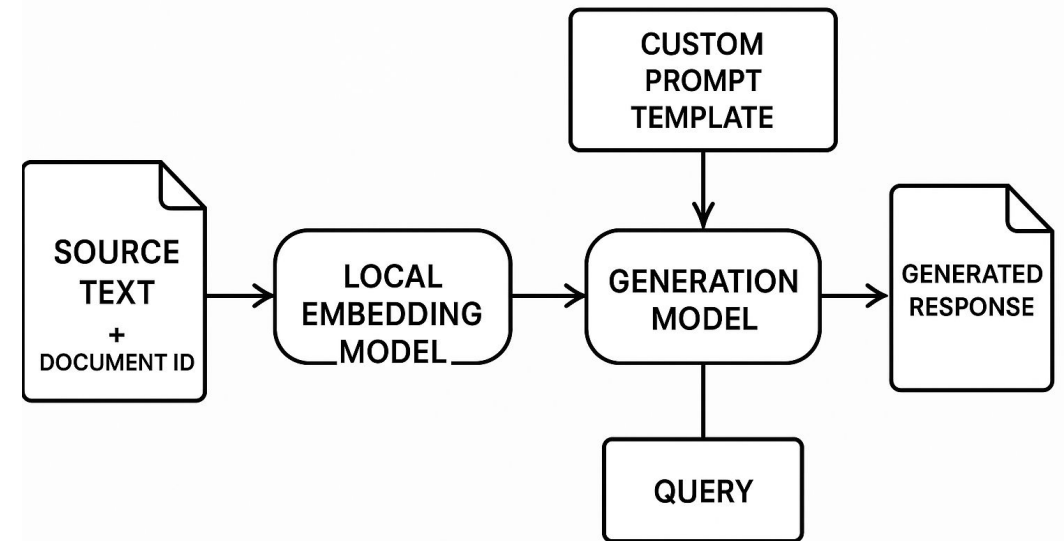
Zero-Shot System

- **No retrieval** component
- **Baseline for the generation pipelines** of the classic and the custom RAG (one-shot) systems
- Prompt **memory**: last 1-2 queries
- Prompt **context**: Query topic
- **Very good results** in terms of semantic similarity to the query and the source text but **hallucinations**



One-Shot System (Custom RAG)

- **Retrieval** based on **semantic search**: source text is transformed into an indexed query engine
- **Retrieved context** from the query engine is matched with the query and imported into a **custom prompt template**
- The output is a **generated response** **conditioned on the source text**, and the document ID
- Similarly good results based on BERT-score but significantly **less hallucination**



Zero- and One-Shot Experiments: Results

Topic	Precision	Recall	F1
kangaroo	0.829427	0.779867	0.803815
Liechtenstein	0.839028	0.777961	0.807242
John Adams	0.837827	0.779302	0.807378
Blaise Pascal	0.845910	0.784550	0.813946
piano	0.826869	0.782122	0.803822
London	0.832180	0.763210	0.796178
English language	0.827927	0.780535	0.803411
Pablo Picasso	0.851256	0.779301	0.813586

Table 1: Zero-shot generation system: Precision, recall, and F1 score results per topic.

Topic	Precision	Recall	F1
kangaroo	0.825054	0.780521	0.802115
Liechtenstein	0.835141	0.779840	0.806395
John Adams	0.828192	0.778246	0.802384
Blaise Pascal	0.838913	0.781415	0.809081
piano	0.824180	0.780222	0.801549
London	0.833670	0.774066	0.802727
English language	0.815381	0.785971	0.800239
Pablo Picasso	0.833290	0.781264	0.806388

Table 1: One-Shot Generation System: Precision, recall, and F1 score results per topic.



Results & Discussion

Retrieval	TF-IDF	Chroma
Precision@1	83.4%	76%
Precision@2	87.2%	84.3%

	Zero-Shot	One-Shot	RAG
Precision	0.8363029814	0.829227536	0.8192106634
Recall	0.7783558304	0.780193089	0.6886122508
F1	0.8061723536	0.8038597454	0.7472603923

- BERTScore not the best evaluation method for this task
- Wrong answers and hallucinations go unnoticed
- Ethical considerations with QA-systems
- Lack of multishot-testing



Error Analysis

Topic	Query	Answer	Retrieved	Gold	Error
Piano	Did Bartolomeo Cristofori invent the modern piano?	Bartolomeo Cristofori did not invent the modern piano [..]	S10_set2_a1	S10_set2_a1	False Positive
London	Over how many languages are spoken in London?	London is home to over 200 languages spoken, with the largest number of native speakers being spoken in Bengali (147,000), followed by Urdu (115,000), Hindi (105,000), and Tamil (95,000).	S09_set5_a1 (English_lang)	S09_set3_a1	False Negative Correctly answered despite wrong document.
kangaroo	What is a roo?	A roo (pronounced "roo-oo") is a type of guitar pick used for strumming, plucking, or picking.	S10_set2_a7 (Guitar)	S08_set1_a1	Wrong answer despite context.
English_l anguage	What is given for the number of native speakers?	The number of native speakers is given for each country in the text material [...]	S09_set5_a1	S09_set5_a1	Vague questions. Gold answer: "No figure is given for the number of native speakers."

Conclusion

- Document retrieval: TF-IDF baseline retrieved documents better than Chroma
- Answer generation: No distinct difference between the RAG with database and Zero-/One-Shot
- Further tests needed; different embedding model with chroma, combining evaluation methods, generating answers with larger models





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Thank you!

Questions?